

SUYASH SREEKUMAR

Software Engineer

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SUMMARY

- Senior Software Engineer with 5+ years of experience building scalable, fault-tolerant backend systems across financial services and enterprise platforms.
- Strong focus on Node.js, TypeScript, and Python, delivering microservices, distributed systems, and event-driven architectures in high-throughput environments.
- Proven impact on improving system performance, latency, and reliability for trading and payment workflows handling thousands of transactions per minute.
- Hands-on experience with system design, cloud-native engineering, and Kafka-based pipelines supporting mission-critical services.
- Built and integrated AI-driven automation, including embedding-based services, vector search, and RAG pipelines to enhance analytics and decision-making.

SKILLS

Backend & Languages: TypeScript, JavaScript (Node.js, Express.js), Python (FastAPI), REST APIs, gRPC, Java, Spring Boot (legacy systems & interoperability), Secure API Design, Secrets Management

System Design & Architecture: Microservices Architecture, Distributed Systems, Event-Driven Design, Kafka Pipelines, Schema Evolution (Avro/Protobuf), Message Ordering, Dead Letter Queues (DLQ), High-Throughput Systems, Scalability & Capacity Planning, Fault Tolerance, Circuit Breakers, Retries & Timeouts, Data Consistency, Idempotent Consumers, Backpressure Handling

Databases & Infrastructure: PostgreSQL, MongoDB, DynamoDB, Redis, Advanced SQL Optimization, AWS (EC2, EKS, RDS, S3, IAM), Docker, Kubernetes, Auto Scaling, Load Balancing, Blue-Green / Canary Deployments, Infrastructure as Code, CI/CD (GitHub Actions, Jenkins, ArgoCD)

Frontend, Observability & AI: React, Next.js, TypeScript, Tailwind CSS, Prometheus, Grafana, Centralized Logging, Alerting, SLOs/SLIs, Generative AI, Vector Databases (FAISS, Pinecone), RAG Pipelines

WORK EXPERIENCE

Black Rock Jan 2025– Present | United States

Senior Software Engineer

- Led modernization of Aladdin backend services by designing Node.js and TypeScript microservices, introducing Kafka-driven async workflows and gRPC communication, expanding peak event-handling capacity by 90K+ messages per risk cycle.
- Optimized data-access paths across portfolio, exposure, and market-data APIs through PostgreSQL and MongoDB schema tuning, Redis caching, and pool stabilization, cutting query latency by 30–60 ms.
- Delivered full-stack features for the Portfolio Insights & Monitoring console using React, Next.js, TypeScript, and Tailwind CSS, enabling analysts to navigate thousands of intraday alerts smoothly.
- Containerized backend compute services with Docker and deployed onto AWS EKS, externalizing secrets via AWS Secrets Manager and SSM, eliminating ~40 manual updates per release cycle.
- Architected distributed trade-lifecycle processing using Kafka, Protobuf schemas, and idempotent consumer groups to maintain stability during volatile trading windows.
- Built an embedding-based anomaly-detection service using Python and vector stores (FAISS/Pinecone), automating the review of 120K+ portfolio signals.
- Reinforced platform security by enforcing OAuth2/JWT patterns and expanding audit-trace coverage aligned with enterprise governance.
- Accelerated delivery velocity by enhancing CI/CD pipelines with GitHub Actions, Jenkins, and ArgoCD, shortening release cycles by several hours per deployment.

Goldman Sachs Jul 2019 – Aug 2023 | India

Software Engineer

- Engineered distributed backend systems for trading, payments, and compliance workflows, scaling throughput from 1.2k to ~1.9k orders per minute during peak market hours.
- Rebuilt real-time event pipelines using Kafka with improved partition balancing, retry strategies, and consumer coordination, supporting 35K+ additional events per hour.
- Developed and maintained backend services across Node.js and Java, ensuring clean API contracts, versioning, and secure integrations with analytics platforms.
- Stabilized JVM-based fraud and risk engines through targeted GC tuning, circuit breakers, and executor-pool optimization, compressing response times from ~420 ms to under 260 ms.
- Migrated legacy FIX routing modules into a Kafka-based microservice architecture, enabling parallel consumers and shortening routing latency by ~110 ms.
- Lowered UI interaction delays for settlement and payments tools by refining React lifecycle patterns, minimizing redundant renders, and applying route-level caching (180–220 ms gains).
- Established multi-region readiness for ledger and audit services using AWS EC2, RDS, S3, IAM, and CloudWatch with blue-green deployments and automated failover checks.
- Expanded observability with Prometheus metrics, trace IDs, and Grafana dashboards, cutting incident investigation time by over 2×.
- Streamlined CI/CD workflows by containerizing services with Docker and restructuring Jenkins pipelines to improve rollout stability.

EDUCATION

Master of Science in Computer Science

Oregon State University, Concentration: Artificial Intelligence, GPA: 3.96/4.00

- Graduate Teaching Assistant - Oregon State University